

Comparative determinants of 4-year cardiovascular event rates in stable outpatients at risk of or with atherothrombosis.

SCIFACT-EVIDENCE-16760369

Evidence abstract

CONTEXT Clinicians and trialists have difficulty with identifying which patients are highest risk for cardiovascular events.

Prior ischemic events, polyvascular disease, and diabetes mellitus have all been identified as predictors of ischemic events, but their comparative contributions to future risk remain unclear.

OBJECTIVE To categorize the risk of cardiovascular events in stable outpatients with various initial manifestations of atherothrombosis using simple clinical descriptors.

DESIGN, SETTING, AND PATIENTS Outpatients with coronary artery disease, cerebrovascular disease, or peripheral arterial disease or with multiple risk factors for atherothrombosis were enrolled in the global Reduction of Atherothrombosis for Continued Health (REACH) Registry and were followed up for as long as 4 years.

Patients from 3647 centers in 29 countries were enrolled between 2003 and 2004 and followed up until 2008.

Final database lock was in April 2009.

MAIN OUTCOME MEASURES Rates of cardiovascular death, myocardial infarction, and stroke.

RESULTS A total of 45,227 patients with baseline data were included in this 4-year analysis.

During the follow-up period, a total of 5481 patients experienced at least 1 event, including 2315 with cardiovascular death, 1228 with myocardial infarction, 1898 with stroke, and 40 with both a myocardial infarction and stroke on the same day.

Among patients with atherothrombosis, those with a prior history of ischemic events at baseline ($n = 21,890$) had the highest rate of subsequent ischemic events (18.3%; 95% confidence interval [CI], 17.4%-19.1%); patients with stable coronary, cerebrovascular, or peripheral artery disease ($n = 15,264$) had a lower risk (12.2%; 95% CI, 11.4%-12.9%); and patients without established atherothrombosis but with risk factors only ($n = 8073$) had the lowest risk (9.1%; 95% CI, 8.3%-9.9%) ($P < .001$ for all comparisons).

In addition, in multivariable modeling, the presence of diabetes (hazard ratio [HR], 1.44; 95% CI, 1.36-1.53; $P < .001$), an ischemic event in the previous year (HR, 1.71; 95% CI, 1.57-1.85; $P < .001$), and polyvascular disease (HR, 1.99; 95% CI, 1.78-2.24; $P < .001$) each were associated with a significantly higher risk of the primary end point.

CONCLUSION Clinical descriptors can assist clinicians in identifying high-risk patients within the broad range of risk for outpatients with atherothrombosis.

Document metadata

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Evidence checklist

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various initial manifestations of atherothrombosis using simple clinical descriptors.

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